

The hidden potential of recovery: how nutrition and supplements change the rehabilitation process

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Abstract

Nutritional rehabilitation is a critical but often overlooked element in clinical practice in the treatment of musculoskeletal injuries and postoperative recovery. This review analyzes the importance of nutritional support in overcoming the hypermetabolic response and anabolic resistance that occur as a result of immobilization. Optimizing caloric intake through an individualized stress factor and ensuring high protein intake (2.0–3.0 g/kg/bm) distributed evenly throughout the day are key strategies for minimizing muscle atrophy. The article outlines the role of specific nutraceuticals such as beta-hydroxy-beta-methylbutyrate (HMB), creatine monohydrate, omega-3 fatty acids (4,000 mg/day), and free forms of essential amino acids for stimulating protein synthesis and bone remodeling. The additional inclusion of vitamin D, antioxidants (vitamins A, C, and E), and probiotics optimizes immune function and tissue healing. Nutrient timing relative to therapeutic sessions is defined as a decisive factor in accelerating functional recovery. The integration of these dietary guidelines into the interdisciplinary care model is essential for a faster and safer return of patients to physical activity.

Keywords

creatine, HMB, immobilization, muscle atrophy, nutritional rehabilitation, omega-3, protein synthesis, vitamin D

Introduction

Nutritional therapy often remains a neglected component in standard rehabilitation protocols, even though trauma and surgical interventions cause a significant hypermetabolic and catabolic response. Optimizing nutritional status is crucial for minimizing muscle atrophy, which begins within the first 36–48 hours of immobilization. It is emphasized that physical activity, in conjunction with optimal nutrition, constitutes a key determinant in the maintenance of a healthy lifestyle (Mihaylova-Alakidi 2019). This review aims to systematize the scientific evidence on

the effectiveness of specific dietary strategies and nutraceuticals (HMB, creatine, omega-3, amino acids) for accelerating the healing process and improving functional outcomes in trauma patients.

The importance of nutritional rehabilitation after traumas

Nutritional therapy (diet therapy, nutritional rehabilitation) is often not recognized as a standard component of rehabilitation interventions. Nutrition and nutritional rehabilitation

have the potential to be accessible, wide-ranging strategies that complement the existing standard of care for traumas, whether treated surgically or conservatively, while also taking into account recovery and rehabilitation (Harlan et al. 1990). Nutritional interventions can accelerate the recovery process and support full treatment; therefore, the inclusion of nutritional strategies is important at every stage of the treatment process. Preoperative nutrition and nutritional needs during rehabilitation are key factors that must be taken into account. Improving the physiological response to wounds, immobilization, and traumatic brain injury can be achieved by optimizing macronutrient composition, calorie intake, nutrient timing, and the inclusion of specific nutritional supplements. Previous studies provide scientific evidence for effective nutritional recommendations aimed at reducing surgical complications, minimizing deficits after immobilization, and increasing the likelihood of a safe return to sports participation in athletes (Howard-Wilsher et al. 2016). Recommendations include determining individual calorie requirements to ensure energy needs are met, increasing protein intake with a focus on evenly distributing it throughout the day to help reduce muscle mass and strength loss during immobilization, effective use of nutritional supplements to overcome difficulties related to adequate calorie intake and reduced appetite, and appropriate timing of intake. The rehabilitation process requires a stable nutritional strategy (nutritional rehabilitation) to improve recovery from injury, involving a team of specialists (sports traumatologist, physiotherapist, kinesiologist, specialist

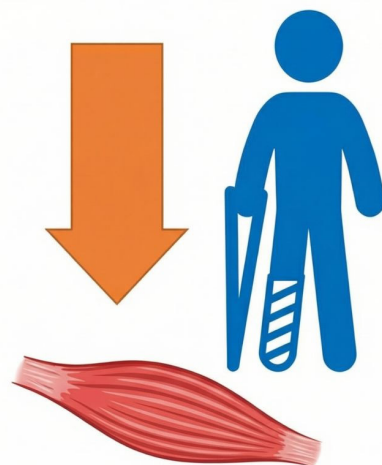
in nutrition and dietetics or dietitian, coaches), and it should be included in standard post-trauma treatment practice. The patient-centered approach is a decisive factor in treatment outcomes and represents an essential practical model for the management and optimization of healthcare, as well as for health promotion (Mihaylova and Lyochkova 2012).

Sports injuries lead to serious financial, mental, and physical problems among patients, which require an interdisciplinary and effective approach to treatment (Harlan et al. 1990; Kantchev et al. 2024). Nutritional interventions are rarely integrated as a standard component of rehabilitation practices, despite their accessibility and proven potential to adequately and effectively complement standard treatment and rehabilitation of the musculoskeletal system.

Injuries are a natural and likely consequence of physical activity. The inflammatory response that follows injury is the initial stage of the recovery process, which can last from several hours to several days, depending on the nature of the injury (Tipton 2015). Since this inflammatory response is essential for the initiation of healing, interventions to reduce inflammation through nutrition during this acute phase may be counterproductive to recovery. However, a rational approach to reducing prolonged or excessive inflammation through nutritional supplementation may accelerate the recovery process and reduce the time to return to sports activities without risk to health. Patients with injuries that require immobilization or surgical intervention deserve special attention in terms of nutrition due to the significant physiological requirements associated with the healing process (Fig. 1)

Key Concerns in Injury & Immobilization

Muscle atrophy
Reduced muscle protein synthesis
Development of anabolic resistance
Proteolysis
Loss of strength



Nutritional Targets

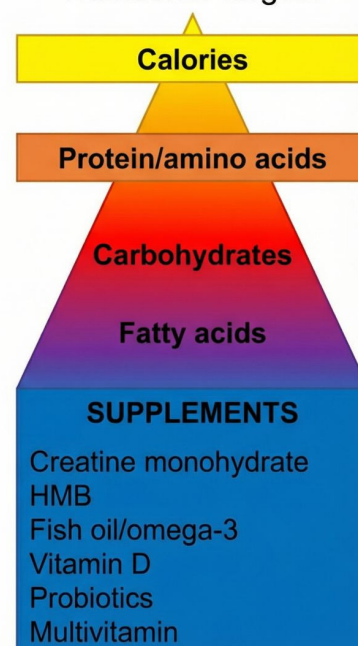


Figure 1. Physiological and metabolic needs after trauma can be met with basic nutritional goals: increased caloric needs, increased protein intake, regulation of blood sugar levels with complex carbohydrates, and essential fatty acids. Furthermore, empirical data suggest that certain nutritional supplements have the potential to alleviate the adverse effects associated with surgical intervention and immobilization, thereby promoting accelerated recovery. Abbreviation: HMB, beta-hydroxy-beta-methylbutyrate (Abbie et al. 2020).

(Abbie et al. 2020). Special attention to nutrition (nutritional rehabilitation) is required regardless of the severity of the injury (minor injuries or contusions, such as muscle strains, stress fractures, or ankle sprains, as well as the separate category of traumatic brain injuries (TBI)), the type of treatment (surgical or conservative), and the phase of treatment: pre-operative, postoperative, or recovery phase.

Preoperative nutritional therapy

When trauma occurs, a cascade of inflammatory, immune, and metabolic responses is initiated, leading to a hypermetabolic state. To support this increased metabolic activity and facilitate the healing process, a significant intake of macro- and micronutrients is required (Bannister and Sattilaro 1962). However, in standard preoperative protocols, patients' nutritional needs prior to surgery are often overlooked; preoperative care typically emphasizes fasting and fluid restriction beginning at midnight on the day of surgery in order to minimize the risk of pulmonary aspiration. In recent years, a growing body of research and clinical evidence suggests that it may be safe for patients to follow less restrictive fasting protocols. It has been shown that the intake of a high-carbohydrate drink by patients immediately before surgery is safe, reduces the catabolic stress response, and potentially improves postoperative outcomes (Ljungqvist and Soreide 2003). The latest guidelines recommend that patients refrain from solid food for at least six hours and clear liquids for at least two hours before surgery. Evidence suggests that maintaining glycemic control before surgery, particularly through carbohydrate intake, improves surgical outcomes by reducing the catabolic response and

insulin resistance associated with surgical stress. Specifically, oral intake of 100 g of glucose the evening before surgery, followed by 50 g two to three hours before the procedure, effectively reduces postoperative insulin resistance and muscle loss after surgery (Smedley et al. 2004). Perioperative—but not pre- and postoperative—administration of an oral supplement (providing 1.5 kcal and 0.05 g of protein per milliliter) reduces postoperative weight loss and complications (Fig. 2) (Abbie et al. 2020).

Postoperative nutritional rehabilitation

In most cases, trauma leads to immobilization or non-weight-bearing of the affected limb. In the absence of activity, muscle mass can decrease in just 36 hours, changes in genomic expression are observed after 48 hours, and within five days of immobilization, there is a significant reduction in muscle tissue (Reich et al. 2010). In general, nutrition during rehabilitation should provide sufficient calories and protein to support wound healing and prevent loss of lean body mass and pure muscle mass (Tipton 2015). Other goals include regulating inflammatory and immune responses, improving blood sugar management, and providing essential macro- and micronutrients to optimize the recovery and healing process. During the rehabilitation phase, the stress response and increased catabolic processes cause an overall increase in energy, protein, amino acid, and micronutrient requirements, which necessitates the first stage of nutritional rehabilitation: determining the appropriate requirements (Fig. 3) (Bannister and Sattilaro 1962; Abbie et al. 2020).

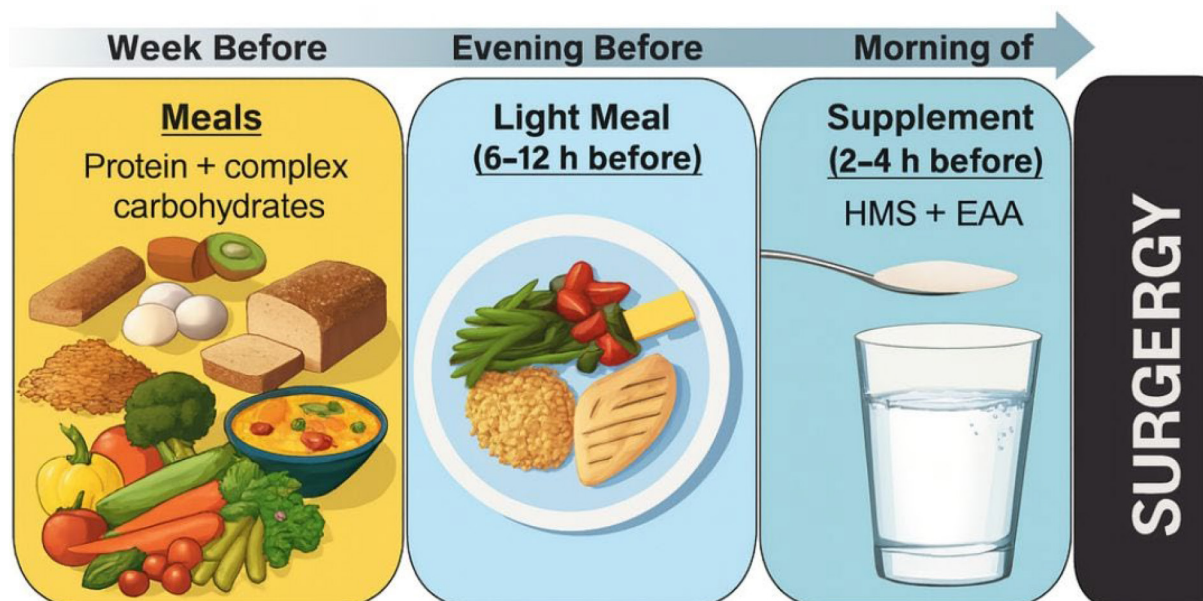


Figure 2. Before surgery, it is recommended to increase the intake of carbohydrates in the diet, especially from complex sources, along with adequate protein intake. The evening meal before surgery should be balanced, including slow-digesting carbohydrates and proteins. In addition, it is recommended to consume a clear liquid containing high molecular weight (HMW) starch and essential amino acids (EAAs) two to four hours before surgery (Abbie et al. 2020).

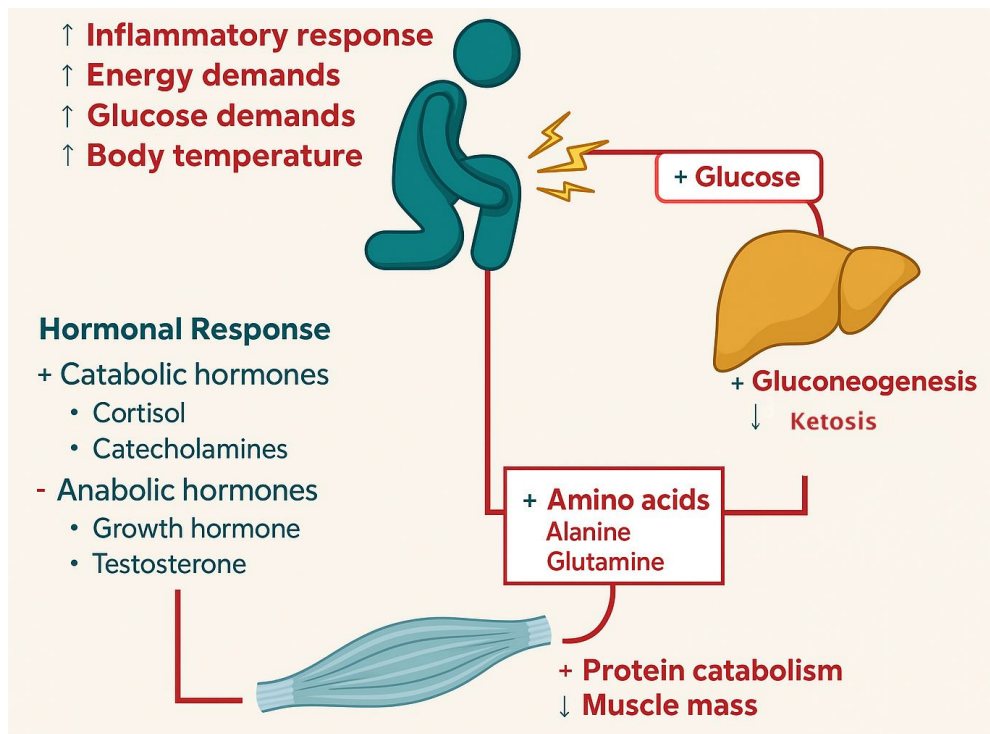


Figure 3. Trauma and surgical intervention provoke a serious stress response, which emphasizes the importance of nutrition as a supportive factor. This response increases the body's need for glucose and amino acids, which in turn stimulates a hormonal response that leads to a catabolic environment, causing a large loss of lean body mass. Adapted from Demling (Abbie et al. 2020)

Calorie needs

Regardless of whether the trauma is treated surgically or conservatively, the patient's basal metabolic rate (BMR) increases to cope with the increased cellular metabolism associated with the healing process (Calder 2013). To determine an individual's calorie needs, it is necessary either to measure the basal metabolic rate (BMR) through indirect calorimetry or to estimate it using predictive equations that take into account the energy required to maintain basic physiological functions and homeostasis (Burke et al. 2019). A commonly used alternative estimate for BMR in young people is approximately 25 kilocalories per kilogram of ideal body weight. During rehabilitation, the metabolic rate increases and must be adjusted through additional caloric intake, which can be estimated by applying a stress factor (Table 1). This specific stress factor can significantly increase the body's metabolic energy needs—from approximately 20% in cases of minor injuries and operations to 100% in cases of more severe physical injuries, such as burns. Trauma to the musculoskeletal system does not usually significantly increase basal energy needs, but in order to prevent stress metabolism, the dominant breakdown of proteins, and reduced physical activity, adequate and increased calorie intake is necessary according to the formula: Total daily energy needs = Basal metabolic rate (BMR) × Stress factor × Physical activity coefficient (Bannister and Sattilaro 1962).

Maintaining caloric balance and meeting increased energy needs after trauma are crucial. Due to reduced physical activity after surgical interventions (e.g., anterior

cruciate ligament, rotator cuff, joint procedures, or fractures) or trauma, many people initially respond by reducing their calorie intake. Although it may seem logical to restrict calories to prevent unwanted weight or fat gain, this can actually be counterproductive. A negative energy balance is likely to worsen recovery outcomes by impeding wound healing and increasing muscle loss (Harris and Benedict 1918). In turn, a negative energy balance is likely to result in increased loss of functional strength and a longer recovery period before returning to active physical and sports activities (Bannister and Sattilaro 1962). Concerns about weight gain and body composition changes due to fat accumulation after injury and during recovery are justified, which requires not only a mechanical increase in calorie intake but also an adjustment of the macronutrient composition of the diet, which plays a decisive role in managing changes in body composition (Dimitrova et al. 2014; Aragon et al. 2017). A diet focused on reducing carbohydrate intake (approximately 40 E% of energy) or maintaining a carbohydrate-to-protein ratio of 2:1 (e.g., 260 g of carbohydrates and 130 g of protein) has been shown to promote favorable changes in body composition. By increasing the proportion of protein intake, along with complex carbohydrates, patients can minimize weight gain at the expense of fat mass while meeting the nutritional needs necessary for an adequate rehabilitation process (Table 1). Alcohol intake should be avoided due to the unnecessary intake of additional energy, its negative effect on protein muscle synthesis, and its inhibition of the wound healing process by suppressing the inflammatory response resulting from trauma (Parr et al. 2014).

Table 1. Determining energy needs during trauma and rehabilitation is a key initial step. Metabolic needs increase to support basic metabolic processes and the increased energy expenditure associated with wound healing and trauma treatment. After surgery and throughout the recovery process, it is important to assess both energy requirements and the quality of calories consumed.

Category	Subcategory description	Value/formula
Basal metabolic rate (RMR)	Men (Harris–Benedict)	$RMR^* = 66 + (13.7 \times wt \text{ (kg)}) + (5 \times ht \text{ (cm)}) - (6.8 \times age \text{ (y)})^{**}$
	Women (Harris–Benedict)	$RMR = 655 + (9.6 \times wt \text{ (kg)}) + (1.8 \times ht \text{ (cm)}) - (4.7 \times age \text{ (y)})$
	Alternative (Cunningham, 1980)	$RMR = 370 + (21.6 \times FFM^* \text{ (kg)})$
Level of physical activity	Sedentary (little or no exercise, office work at a desk)	1.2
	Lightly active (light exercise/sports 1–3 days/week)	1.375
	Moderately active (moderate exercise/sports 6–7 days/week)	1.55
	Very active (1–2 intense workouts every day)	1.725
	Extremely active (heavy workouts 2+ times/day; prolonged endurance training for competition)	1.9
Stress factor	Mild (minor injury—sprained ankle, dislocation, bone fracture, minor surgery, and clean wound)	1.2
	Severe (infected wound, severe trauma—cruciate ligament surgery and severe burn)	1.5
Determining energy requirements and macronutrients in clinical practice		
MAN with diagnosis: Recovery after rotator cuff injury (21 years old, 1.83 m, 81.6 kg)	Activity level: Sedentary—1.2	RMR: 2000–2045 kcal/day
	Stress factor: 1.2 (minor surgery)	Calculated total energy requirements: 2900–3000 kcal/day Macronutrients: Protein: 2.0–3.0 (upper limit) g/kg/b.w.; (164–245 g/day) Fats: 1.0 g/kg b.w.; (82 g/day) Carbohydrates: 3.0–5.0 g/kg/b.w.; (245–410 g/day)
WOMAN diagnosed with: Recovery after surgery for anterior cruciate ligament rupture (18 years old, 1.63 m, 59.0 kg)	Activity level: Sedentary—1.2	RMR: 1430–1480 kcal/day
	Stress factor: 1.2 (minor surgery)	Calculated total energy requirements: 2000–2100 kcal/day ^{***} Macronutrients: Protein: 2.0–3.0 (upper limit) g/kg/b.w.; (120–180 g/day) Fats: 1.0 g/kg/b.w.; (60 g/day) Carbohydrates: 3.0–5.0 g/kg/b.w.; (180–300 g/day)

* Abbreviations: RMR – resting metabolic rate; FFM – fat-free mass.

** The general formula is $TEE = RMR \times activity\ coefficient \times stress\ factor$.

*** A diet that includes complex carbohydrates, high-quality proteins, and healthy fats is recommended. For example, you can prepare a plate with a 30 g serving of complete proteins combined with a serving of whole grains. In addition, including a variety of colorful fresh fruits and vegetables can provide essential antioxidants that aid recovery, reduce inflammation, and provide vital micronutrients (Abbie et al. 2020).

Recommendations for protein intake

After trauma, the body's need for amino acids increases significantly. This increased requirement for amino acids is essential for facilitating wound healing, repairing damaged tissue, and regulating blood sugar levels (Demling 2009). In the absence of adequate nutritional support, this increased metabolic need is met by the breakdown of skeletal muscle tissue. The degree of muscle loss can vary significantly depending on factors such as the severity of the injury, the degree of immobilization imposed on the patient, and the time between the injury and subsequent surgical intervention. Loss of fat-free mass (FFM) is an independent factor associated with a longer hospital stay.

Inadequate energy and protein intake can hinder healing and recovery of incisions, tissue grafts, and other forms of secondary trauma. Recommendations include increased protein intake preoperatively (several days before surgery) to minimize muscle mass loss and accelerate recovery (Demling 2009).

When a stress response is triggered by trauma or injury, the normal homeostatic mechanisms that work to prevent FFM loss are disrupted. This leads to increased energy expenditure and accelerated protein metabolism (Jäger et al. 2017). Stress triggered by trauma significantly increases the body's protein needs, often by approximately 80% above normal baseline levels. For physically active people, a daily protein intake of between 1.4 and 2.0 g/kg/bm is generally recommended to maintain a neutral nitrogen and protein balance.

During recovery from injury and periods of immobility, to maintain the body's overall protein balance, daily protein intake requirements increase due to increased muscle protein breakdown (Jäger et al. 2017). The nutritional goals during this period should be consistent with traditional anabolic goals, as increased catabolic hormones can be compensated for by increasing protein intake, thus achieving a net protein balance. Therefore, during rehabilitation, the recommended minimum protein intake is increased from 1.6 g/kg/bm/day to 2.0–3.0 g/kg/bm, with an emphasis on the additional inclusion of approximately 3 g of leucine at each meal (Nicastro et al. 2011). Leucine is often referred to as an “anabolic trigger” and is the main amino acid responsible for initiating protein synthesis in muscles. Consuming foods rich in leucine, such as various animal proteins, including chicken, beef, milk, and fish, as well as whey protein, can aid faster recovery from trauma (Nicastro et al. 2011; Tipton 2015).

A dose of 30 g of protein has been found to be optimal for maximizing protein synthesis in healthy individuals. However, the dose of protein required to stimulate protein synthesis during periods of immobility and inactivity, mainly due to the development of anabolic resistance, is likely to increase (Wall et al. 2013). In addition to increased protein needs, the timing of protein intake plays a crucial role in ensuring complete absorption and optimizing protein synthesis. To reduce protein breakdown in muscles, it is recommended that each meal provide between 20 and 40 g of total protein (essential, non-essential, and leucine-rich food sources). Protein intake should occur within one hour of waking, then at three- to four-hour intervals, shortly before rehabilitation, and also before bedtime (Areta et al. 2013).

Complex carbohydrates

Carbohydrates are a primary and vital source of energy during rehabilitation. Their contribution to the recovery process is complex (immunological, hormonal regulation, and enzyme activity) (Demling 2009). They play a key role in preserving protein during catabolic states; for example, it has been shown that a diet rich in carbohydrates reduces muscle protein breakdown more effectively than a diet rich in fat (Hart et al. 2001). The recommended energy intake of complex carbohydrates in rehabilitation nutrition after trauma is 3 to 5 g/kg/bm, or approximately 55 E% of total daily calorie intake, including whole grains, fruits, vegetables, and dairy products. As the level of physical activity increases, the body's carbohydrate needs also increase. It is recommended that no more than 60% of the total calorie intake be provided by carbohydrates, because exceeding this threshold can lead to hyperglycemia, which in turn hinders the healing process and has a negative immunomodulatory effect. Simple carbohydrates in the form of processed and refined sugars should be limited in

favor of complex carbohydrates, which help maintain stable blood sugar levels and are also a good source of essential vitamins, minerals, and fiber.

Essential fatty acids

An inflammatory response is triggered immediately after a traumatic event and plays a vital role in ensuring effective treatment, but if it is prolonged, which is often observed after severe trauma or surgery, it can be counterproductive and hinder the healing process (Stechmiller 2010). That is why adequate nutritional intake of essential fatty acids is necessary in rehabilitation nutrition to modulate and suppress the intensity of the prolonged inflammatory response. Fats are a major source of energy that aids wound healing and promotes cell growth. Polyunsaturated fatty acids (PUFAs)—alpha-linolenic acid (omega-3 fatty acid) and linoleic acid (omega-6 fatty acid)—and monounsaturated fatty acids (MUFA)—oleic acid—are extremely important for optimal recovery through their role in the formation of cell membranes, while saturated fats are used primarily as a source of energy. The recommendations for fat intake are as follows: 20% to 25% E% of the total calorie intake should come from fat, or 0.8 to 2 g/kg/body weight/day (due to calorie density), of which omega-3 PUFA are approximately 2 g/day and omega-6 PUFA are approximately 10 g/day (Calder 2013). Foods that are rich sources of fat include avocados, olive oil, various types of fish, flaxseed, nuts, and seeds. During the recovery phase, it is recommended to limit the consumption of omega-6 PUFAs, whose sources include vegetable oils, processed meats, fried and breaded foods, and high-fat foods, due to their pro-inflammatory properties (especially in cases of an imbalance in the omega-3/omega-6 PUFA ratio and high intake of arachidonic fatty acid and TFA).

Recommendations regarding diet

It is believed that the key to supporting recovery in the rehabilitation process is an adequate diet by distributing energy and nutrient intake throughout the day, rather than simply counting total daily consumption (Aragon and Schoenfeld 2013). There is evidence that timing nutrient intake plays an important role in stimulating protein synthesis, reducing muscle protein breakdown, promoting recovery, and improving body composition (Schoenfeld et al. 2013). With regard to injuries and recovery, the diet should be coordinated with the rehabilitation process, as it can significantly affect changes in lean muscle mass (LMM), muscle strength and function, as well as the athlete's return to physical activity. The hours and minutes immediately before and after therapeutic rehabilitation sessions play a crucial role in ensuring access to carbohydrates and proteins—essential nutrients that support muscle activity during training and therapy—which also improves results (Fig. 4) (Kerksick et al. 2017; Abbie et al. 2020).

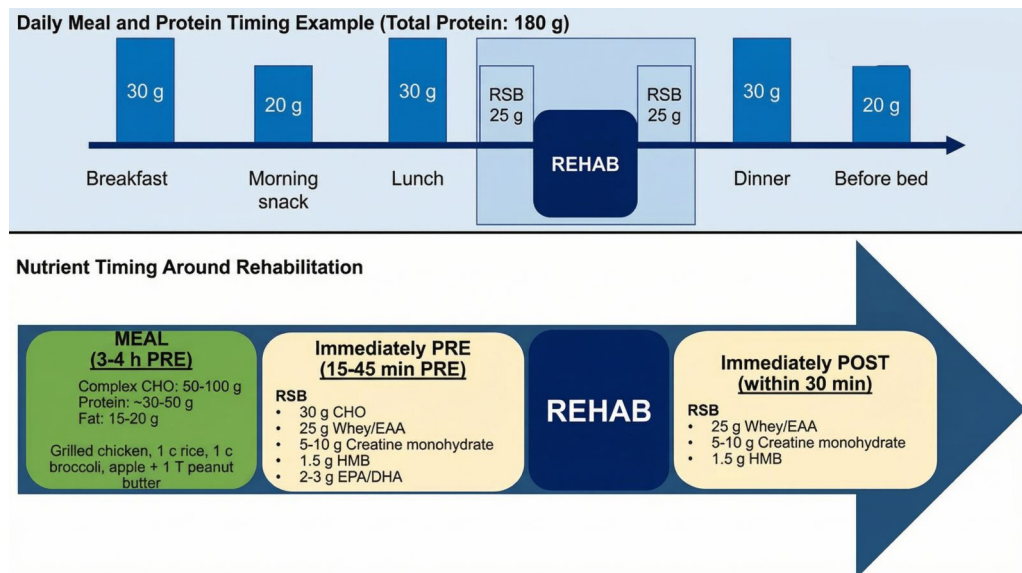


Figure 4. Planning macronutrient intake throughout the day is essential, especially with regard to nutritional intake before and after therapeutic interventions. Abbreviations include EAA for essential amino acids, HMB for beta-hydroxy-beta-methylbutyrate, and RSB for a combination of rehabilitation supplements (Abbie et al. 2020).

Nutritional supplements

Improving the quality and timing of nutrient intake, along with the inclusion of selected nutritional supplements, can further enhance the recovery process and shorten the time needed to return to adequate physical and athletic activity. Several nutritional supplements have proven effective in supporting tissue recovery and regeneration, helping to reduce muscle loss, especially when taken after surgery or during periods of immobility (Ivanova et al. 2016).

Creatine monohydrate

Creatine is an organic compound produced endogenously in limited quantities from the amino acids arginine, methionine, and glycine and through the intake of protein-rich foods (fish, beef, and other meats). Approximately 95% of all creatine stores in the body are stored in skeletal muscles (Buford et al. 2007). Many researchers have shown that taking creatine in the form of supplements leads to an increase in its levels. Creatine monohydrate (CrM) has established itself as one of the most effective ergogenic aids for improving performance during high-intensity training and for stimulating the accumulation of FFM and LMM in combination with adequate physical activity (Candow and Chilibeck 2010). In addition, creatine supplementation has the potential to reduce the severity and frequency of traumatic brain injuries by maintaining neuromuscular health and improving muscle function and coordination, improving bone mineral density (BMD), hydration, and thermoregulation processes, which in turn modulate the risk of muscle cramps and dehydration (Sobolewski et al. 2011). Creatine intake (a supplement of 20 g CrM (4 × 5 g) during periods of immobility) has shown inconsistent results on muscle size and strength in various studies

with and without an added rehabilitation program. When CrM supplementation is combined with rehabilitation, it has been found to significantly improve muscle mass recovery and hypertrophy in all muscle fiber types compared to the group without CrM intake. Although the exact mechanisms by which CrM may be beneficial to immobilized muscle have not been established, some data suggest that this is a result of increased regulation of myogenic transcription factors (myogenin) and satellite muscle cell activity. With regard to skeletal injuries, CrM may improve bone remodeling by stimulating cell growth, differentiation, and mineralization in bone tissue (Johnston et al. 2009). Creatine monohydrate is the form of creatine that has been most extensively studied in scientific research and is considered the most bioavailable and effective option, with no adverse effects associated with CrM supplementation reported to date. The only side effect reported in some people is slight weight gain, which usually subsides within one to two days after starting CrM supplementation. After immobilization or trauma, it is recommended to start with a loading dose of 20 g (4 × 5 g per day) for a period of five days, then switch to a maintenance dose of 3 to 5 g per day to maintain elevated levels (Hespel et al. 2001; Kreider et al. 2017).

Omega-3 (n-3) polyunsaturated fatty acids

The potential benefits of long-chain omega-3 fatty acids, particularly eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), have been studied in the context of recovery from trauma. These nutrients are classified as essential because, although EPA and DHA can be synthesized from alpha-linolenic acid (ALA), humans do not have the ability to synthesize ALA themselves.

This requires EPA and DHA to be obtained directly from dietary sources. Only fish and fish oils contain significant amounts of EPA and DHA, which are integrated into the cell membranes of every tissue in the body. It is important to note that plant sources of omega-3 (via alpha-linolenic acid) have a very limited ability to be converted into EPA and DHA, making them ineffective sources of omega-3. Omega-3 fatty acids are widely recognized for their anti-inflammatory properties in various chronic inflammatory conditions (Calder et al. 2009). However, during the acute phase of trauma or surgical intervention, it is important to carefully consider the role of inflammation, as the body's innate inflammatory response plays a key role in the healing process. Therefore, the administration of high doses of omega-3 fatty acids shortly after trauma, or surgery—within a few hours or just a few days—is not always necessary or justified. In cases where inflammation lasts too long or is difficult to control, including omega-3 fatty acids in the diet can serve as a successful strategy to help control the inflammatory response, especially in immobile patients, by activating muscle protein synthesis and reducing or preventing muscle atrophy and loss. Daily intake of 4,000 mg of omega-3 fatty acids stimulates protein synthesis in muscles, especially in response to increased protein and insulin levels in healthy young people (Smith et al. 2011). In conditions following trauma and immobilization, fish oil supplementation could attenuate muscle loss, although the available publications have not consistently demonstrated these benefits, and there is still no clear consensus or standardized guidelines on the appropriate dosage or application strategies to achieve such effects (Abbie et al. 2020). Taking a supplement of between 2,000 and 4,000 mg of fish oil may help reduce chronic inflammation in the body and improve protein synthesis, bearing in mind that thrombocytopenia and, consequently, an increased risk of bleeding are potential side effects of fish oil supplementation. Current thinking is that reduced platelet activity does not lead to an increased risk of bleeding during or after surgical procedures, and it is generally considered unnecessary to discontinue supplementation (Begtrup et al. 2017). Assessing the quality of fish oil is essential to ensure that contamination from elevated levels of mercury is minimized.

Essential amino acids

Protein supplementation is an affordable and effective way to ensure adequate protein intake. Although it is important to prioritize high-quality, complete dietary sources of protein, protein supplements offer a more concentrated and direct source of amino acids compared to regular dietary protein, resulting in a more powerful anabolic response in the body (Paddon-Jones et al. 2005). When evaluating the quality of a protein source, it is important to consider several key factors related to muscle protein synthesis, namely the percentage of protein as essential amino acids (EAAs) and leucine content,

as well as bioavailability and digestibility. Whey protein is considered an excellent source of high-quality protein, containing over 50% of its amino acid profile derived from essential amino acids (EAAs) and providing approximately 2.7 g of leucine per standard serving (25 g of protein) (Van Vliet et al. 2015). Among the several types of whey protein, the most common are isolate and concentrate. Due to the higher level of filtration that isolate undergoes compared to concentrate, it has a higher protein content per serving and contains very little or no fat and lactose. As a result, it is more suitable for people with lactose intolerance. Whey protein is a rapidly digestible source of protein that quickly raises the levels of essential amino acids (EAAs) in the blood. This rapid absorption makes it particularly suitable as a protein supplement both before and after physical activity and during rehabilitation. Unlike other proteins, casein, which is extracted from fresh milk, is absorbed much more slowly and is therefore recommended to be taken before bedtime in a dose of 30 to 40 g, which can significantly increase muscle protein synthesis (Res et al. 2012). Plant-based protein products have been gaining popularity lately. Among them, soy and pea proteins are considered the highest quality and most preferred due to their rich amino acid profile and effectiveness. In addition, blends that include hemp, pumpkin, and rice proteins are also often used to diversify protein intake and increase nutritional value. However, it is important to note that the rate of muscle protein synthesis stimulated by plant proteins is generally significantly lower than that of animal protein sources, even when plant and animal proteins are comparable in nitrogen and caloric energy content. This reduced response to stimulation stems from lower concentrations of essential amino acids (EAAs) and leucine. Increasing plant protein consumption may improve its ability to support muscle growth and recovery, known as its anabolic potential. For example, many plant protein supplements typically provide approximately 20 g of protein. However, to achieve a leucine intake equivalent to that found in 25 g of whey protein, which provides approximately 2.7 g of leucine, a person would need to consume approximately 38 g of pea protein, 40 g of soy protein, or 54 g of hemp protein (Phillips 2012).

Free-form NAC supplements are also available, containing only NAC (without other macronutrients) in a form that can be rapidly absorbed and used directly by the muscles, improving the anabolic response compared to supplements containing dietary or whey proteins (Paddon-Jones et al. 2005). Taking a supplement consisting of 16.5 g of NAC along with 30 g of carbohydrates proved to be an effective formula for preserving lean leg muscle mass and minimizing strength loss in healthy men after a 28-day period of bed rest (Paddon-Jones et al. 2004). Taking 20 g of NAC twice daily for 1 week before and 2 weeks after total knee replacement surgery reduced the degree of muscle atrophy (in the quadriceps and hamstrings) and showed earlier recovery of functional mobility within 6 weeks after surgery compared to

placebo (Dreyer et al. 2013). Taking NAC immediately before training may improve muscle protein synthesis more effectively than supplementing with it after training, due to the greater amount of amino acids delivered to the muscles (Dreyer et al. 2013). In addition, taking NAC immediately before surgery or therapeutic procedures has been shown to offer significant benefits.

Beta-hydroxy-beta-methylbutyrate (HMB)

HMB is a naturally occurring metabolite derived from the essential amino acid leucine that plays an important role in preventing protein breakdown and promoting de novo protein synthesis. It is particularly important during stressful physiological conditions, after trauma or surgical procedures (Nissen and Sharp 2003). HMB has the potential to influence the activity of enzymes involved in the breakdown of muscle tissue. During periods of limited movement or immobilization, HMB can stimulate the process of protein synthesis by activating the mTOR signaling pathway, leading to improved recovery and regeneration of muscle and tendon tissue, while at the same time helping to reduce myofibril breakdown and modulating the rate of muscle mass loss. Taking 3 g (2×1.5 g) of HMB supplements per day compared to a placebo in elderly, immobile patients for 10 days leads to better maintenance of total lean body mass (FFM) (0.17 kg vs. 2.05 kg in the placebo group) and FFM in the legs (0.08 kg vs. 1.01 kg in the placebo group) (Molfinio et al. 2013). Furthermore, in older people, consumption of 3 g of HMB daily for 12 weeks, combined with progressively increasing strength training, led to a significantly greater improvement in FFM and upper body strength compared to other interventions. HMB is found in foods in the normal dietary intake (0.25 to 1 g daily); supplemental intake of 3 g per day, divided into doses of 1.5 g, has shown the most effective results for maintaining muscle mass (Stout et al. 2013). Supplemental HMB intake can preserve FFM during surgery and immobilization, as well as increase it when combined with rehabilitation therapy, with no reported side effects from its use.

Prebiotics and probiotics

Recent scientific data show that the microbiome in the gastrointestinal tract may play a beneficial role in improving the results of nutritional rehabilitation. Approximately 70% of the body's immune system is located in the digestive system, highlighting the important role that the gut microbiota plays in maintaining immune health. Prebiotics, which are a specific category of nutrients derived primarily from dietary fiber, are selectively broken down by the gut microbiota. This fermentation process generates essential nutrients such as short-chain fatty acids, which serve as energy sources for the microbiome itself. In

addition, prebiotics contribute to the stability and survival of probiotic bacteria as they pass through the stomach (Davani-Davari et al. 2019). Probiotics, known as "good" bacteria, are live microorganisms that provide health benefits to the host when consumed in sufficient and consistent amounts. From a surgical perspective, the use of antibiotics often leads to a negative change in the composition of the gastrointestinal microbiota, which can result in undesirable changes in the processing and metabolism of nutrients in the body. The use of probiotics and prebiotics in patients after surgery can reduce the incidence of bacterial infections as a result of surgery by increasing the local immunity of the mucous membranes, which in turn helps to reduce both the duration of antibiotic treatment and the associated side effects (Raves et al. 2005).

The health benefits associated with probiotics vary depending on the specific strains used and the dose administered. Some strains, such as *Lactobacillus acidophilus* and *Bifidobacterium longum*, have been shown to have a beneficial effect on the immune system. In addition, *Bacillus coagulans* has been shown to improve protein absorption, which is important in rehabilitation (Jäger et al. 2018). The recommended amount to be consumed of each strain is $\geq 10^{10}$ colony-forming units, which can be included in fermented foods such as yogurt or taken as dietary supplements. Probiotic supplements are generally more effective when stored in the refrigerator. For optimal results, it is recommended to consume them daily on an empty stomach, with probiotic supplements containing a large number of colony-forming units—particularly strains such as *L. acidophilus* and *B. longum*—being recommended.

Vitamin supplementation

Vitamin D

Vitamin D is widely recognized for its crucial role in regulating calcium levels and maintaining bone mineral health. Its pleiotropic functions are related to effects on the body's innate and adaptive immune responses and the proper functioning of skeletal muscles, suggesting a role for vitamin D in improving and optimizing the body's recovery during the rehabilitation process. Vitamin D deficiency is defined as plasma levels of 25-hydroxyvitamin (25[OH] D) below 75 nmol/L or less than 30 ng/L. Despite ongoing research, there are no universally agreed optimal levels of vitamin D sufficiency, with the Institute of Medicine accepting levels above 50 nmol/L as sufficient (Institute of Medicine 2010). There is evidence to support the hypothesis that individuals with vitamin D insufficiency or deficiency may have impaired muscle recovery, regeneration, and hypertrophy, with low vitamin D levels in particular being associated with prolonged recovery periods after knee surgery (Nissen and Sharp 2003). It is believed that if vitamin D levels are sufficient, no additional supplementation is necessary; in fact, the use of supplements after bone fractures may potentially have an adverse effect, as it

may lead to the suppression of specific macrophages. The results of studies on the effectiveness of vitamin D intake in daily doses of 2,000 to 4,000 IU per day for different periods of time and in different age groups are contradictory in terms of muscle strength recovery, immunomodulatory effect, inflammation biomarkers, prevention of initial inflammation, and oxidative stress after trauma (Abbie et al. 2020). The dose and duration of supplementation, the age of the patients, and the presence of ongoing chronic inflammation, which in itself can lead to and exacerbate vitamin D deficiency in the body, are important factors. During periods when the immune system is suppressed or when a person is experiencing an infection or significant inflammatory reactions, taking vitamin D supplements may help prevent further vitamin D loss. It is important to analyze vitamin D status during a traumatic event to prevent potential adverse effects that may arise from insufficient or deficient levels of the vitamin. The most optimal dose is probably 4,000 IU per day, unless the patient has already suffered a bone fracture (Barker et al. 2011). Ideally, insufficiency or deficiency would be identified prior to trauma. Through nutritional intake from foods rich in vitamin D—cheese, yogurt, various types of fatty fish (salmon or mackerel), meat, egg yolk, or fortified foods such as orange juice and some cereal snacks—it is possible to obtain a daily intake of 100 to 200 IU of the vitamin, with its absorption being identical from food and supplements, provided that dietary fat is present in the food portion.

Vitamins A, E, and C

Vitamins and minerals are among the most commonly used dietary supplements. Unlike other types of supplements, vitamins and certain minerals are considered essential because of their vital role in maintaining normal physiological functions in the body. Deficiencies in most micronutrients can lead to various health problems and should be prevented, especially during periods of increased physical or emotional stress. In addition, during recovery from trauma or surgery, the body's metabolic needs are increased, making it particularly important to pay special attention to micronutrient intake. Vitamins A, C, and E play a key role in immunonutrition, trauma recovery, and wound healing (Carr and Maggini 2017). Vitamin A has a positive effect on the wound healing process even in patients who are not deficient in this vitamin (Raverdeau and Mills 2014). Vitamin C supplementation appears to be most beneficial for patients experiencing severe stress or trauma, while vitamin E supplementation reduces oxidative stress in the body, which in turn can reduce wound healing time (Carr and Maggini 2017). Current recommendations for vitamin and mineral intake are that they should be obtained through a balanced diet, but if the recommended daily intake cannot be achieved, fortified foods are introduced as an additional option, and when additional support is still needed, a multivitamin dietary supplement may be considered (Institute of Medicine 2005; Institute of Medicine (US), Food and Nutrition Board 2017). There is limited

information on the specific dosage and frequency of intake of these supplements to achieve the desired effectiveness. It has not been proven that taking micronutrients above the usual recommended amounts further increases their effectiveness (Tipton 2015). Although there is no direct evidence, the overall assessment indicates that the risk-benefit ratio associated with taking multivitamins is generally low to moderate, with moderate potential benefits and significantly few risks.

Conclusion

Nutrition plays an important role in recovery and rehabilitation after trauma. Extensive scientific data form the basis for practical nutritional recommendations aimed at reducing complications during surgical treatment, minimizing muscle loss during periods of immobilization, and optimizing the likelihood of a successful return to sports activities. First and foremost, it is important to determine each individual's calorie needs to ensure that their energy requirements are adequately met. Ensuring adequate protein intake, with particular attention to its even distribution across all meals throughout the day, plays a key role in minimizing muscle mass and strength loss, especially during periods of immobility. Nutritional supplements can be useful for managing adequate calorie intake and timing of intake, especially in cases of reduced appetite. When integrated into a comprehensive and well-structured nutritional plan, they can enhance and potentially accelerate the results of effective therapy and rehabilitation, thereby facilitating faster and safer recovery for patients and enabling an earlier return to physical activity.

Nutritional rehabilitation should be an integral part of standard clinical practice. Interdisciplinary collaboration between surgeons, physical therapists, kinesiotherapists, and dietitians is essential for the implementation of these strategies, which ensure a faster and safer return to physical activity.

Additional information

Conflict of interest

The authors have declared that no competing interests exist.

Ethical statements

The authors declared that no clinical trials were used in the present study.

The authors declared that no experiments on humans or human tissues were performed for the present study.

The authors declared that no informed consent was obtained from the humans, donors or donors' representatives participating in the study.

The authors declared that no experiments on animals were performed for the present study.

The authors declared that no commercially available immortalised human and animal cell lines were used in the present study.

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Data availability

All of the data that support the findings of this study are available in the main text.

References

- Abbie E, Smith-Ryan AE, Hirsch KR, Saylor HE, Gould LM, Blue MN (2020) Nutritional considerations and strategies to facilitate injury recovery and rehabilitation. *Journal of Athletic Training* 55(9): 918–930. <https://doi.org/10.4085/1062-6050-550-19>
- Aragon AA, Schoenfeld BJ, Wildman R, Kleiner S, VanDusseldorp T, Taylor L, Antonio J (2017) International society of sports nutrition position stand: diets and body composition. *Journal of the International Society of Sports Nutrition* 14: 16. <https://doi.org/10.1186/s12970-017-0174-y>
- Aragon AA, Schoenfeld BJ (2013) Nutrient timing revisited: is there a post-exercise anabolic window? *Journal of the International Society of Sports Nutrition* 10: 5. <https://doi.org/10.1186/1550-2783-10-5>
- Areta JL, Burke LM, Ross ML, Camera DM, West DW, Broad EM, Coffey VG (2013) Timing and distribution of protein ingestion during prolonged recovery from resistance exercise alters myofibrillar protein synthesis. *Journal of Physiology* 591(9): 2319–2331. <https://doi.org/10.1113/jphysiol.2012.244897>
- Bannister WK, Sattilaro AJ (1962) Vomiting and aspiration during anesthesia. *Anesthesiology* 23(2): 251–264. <https://doi.org/10.1097/0000542-196203000-00012>
- Barker T, Martins TB, Hill HR, Kjeldsberg CR, Trawick RH, Weaver LK, Traber MG (2011) Low vitamin D impairs strength recovery after anterior cruciate ligament surgery. *Journal of Evidence-Based Complementary & Alternative Medicine* 16(3): 201–209. <https://doi.org/10.1177/2156587211413768>
- Begtrup KM, Krag AE, Hvas AM (2017) No impact of fish oil supplements on bleeding risk: a systematic review. *Dan Med J* 64(5): A5366.
- Buford TW, Kreider RB, Stout JR, Greenwood M, Campbell B, Spano M, Antonio J (2007) International Society of Sports Nutrition position stand: creatine supplementation and exercise. *Journal of the International Society of Sports Nutrition* 4: 6. <https://doi.org/10.1186/1550-2783-4-6>
- Burke LM, Castell LM, Casa DJ, Close GL, Costa RJ, Desbrow B, Stellingwerff T (2019) International association of athletics federations consensus statement 2019: nutrition for athletics. *International Journal of Sport Nutrition and Exercise Metabolism* 29(2): 73–84. <https://doi.org/10.1123/ijsnem.2019-0065>
- Calder PC, Albers R, Antoine JM, Blum S, Bourdet-Sicard R, Ferns GA, Zhao J (2009) Inflammatory disease processes and interactions with nutrition. *British Journal of Nutrition* 101(S1): 1–45. <https://doi.org/10.1017/S0007114509377867>
- Calder PC (2013) n-3 fatty acids, inflammation and immunity: new mechanisms to explain old actions. *Proceedings of the Nutrition Society* 72(3): 326–336. <https://doi.org/10.1017/S0029665113001031>
- Candow DG, Chilibeck PD (2010) Potential of creatine supplementation for improving aging bone health. *Journal of Nutrition, Health and Aging* 14(2): 149–153. <https://doi.org/10.1007/s12603-009-0224-5>
- Carr AC, Maggini S (2017) Vitamin C and immune function. *Nutrients* 9(11): 1211. <https://doi.org/10.3390/nu9111211>
- Davani-Davari D, Negahdaripour M, Karimzadeh I, Seifan M, Mohkam M, Masoumi SJ, Ghasemi Y (2019) Prebiotics: definition, types, sources, mechanisms, and clinical applications. *Foods* 8(3): 92. <https://doi.org/10.3390/foods8030092>
- Demling RH (2009) Nutrition, anabolism, and the wound healing process: an overview. *Eplasty* 9: e9.
- Dimitrova R, Petkova V, Dimitrov M, Madzharov V, Nikolova I, Petkova E, Andreevska K, Grekova D, Gueurguiev S (2014) Obesity–relationship with vascular dysfunction. *Advances in Obesity, Weight Management & Control* 1(1): 1–5. <https://doi.org/10.15406/aowmc.2014.01.00001>
- Dreyer HC, Strycker LA, Senesac HA, Hocker AD, Smolkowski K, Shah SN, Jewett BA (2013) Essential amino acid supplementation in patients following total knee arthroplasty. *Journal of Clinical Investigation* 123(11): 4654–4666. <https://doi.org/10.1172/JCI70160>
- Harlan LC, Harlan WR, Parsons PE (1990) The economic impact of injuries: a major source of medical costs. *American Journal of Public Health* 80(4): 453–459. <https://doi.org/10.2105/ajph.80.4.453>
- Harris JA, Benedict FG (1918) A biometric study of basal metabolism in man. *Carnegie Institution of Washington, Washington DC. Proceedings of the National Academy of Sciences* 4(12): 370–373. <https://doi.org/10.1073/pnas.4.12.370>
- Hart DW, Wolf SE, Zhang XJ, Chinkes DL, Buffalo MC, Matin SI, Herndon DN (2001) Efficacy of a high-carbohydrate diet in catabolic illness. *Critical Care Medicine* 29(7): 1318–1324. <https://doi.org/10.1097/00003246-200107000-00004>
- Hespeel P, Op't Eijnde B, Van Leemputte M, Ursø B, Greenhaff PL, La-barque V, Richter EA (2001) Oral creatine supplementation facilitates the rehabilitation of disuse atrophy and alters the expression of muscle myogenic factors in humans. *Journal of Physiology* 536(2): 625–633.
- Howard-Wilsher S, Irvine L, Fan H, Shakespeare T, Suhrcke M, Horton S, Song F (2016) Systematic overview of economic evaluations of health-related rehabilitation. *Disability and Health Journal* 9(1): 11–25. <https://doi.org/10.1016/j.dhjo.2015.08.009>
- Institute of Medicine (2005) *Nutrient Composition of Rations for Short-Term, High-Intensity Combat Operations*. National Academies Press, Washington DC. <https://doi.org/10.17226/11325>
- Institute of Medicine (2010) *Dietary Reference Intakes for Calcium and Vitamin D*. National Academy Press, Washington DC.
- Institute of Medicine (US), Food and Nutrition Board (2017) *Dietary Reference Intakes: The Essential Guide to Nutrient Requirements*. National Academies Press, Washington DC. <https://doi.org/10.17226/11537>
- Ivanova S, Ivanov K, Mladenov R, Papanov S, Ivanova S, Obreshkova D, Atanasov R, Petkova V (2016) Food supplements with anabolic and androgenic activity – UHPLC analysis of food additives containing

- Tribulus terrestris extract. *World Journal of Pharmaceutical Research* 5(3): 6–13.
- Jäger R, Kerksick CM, Campbell BI, Cribb PJ, Wells SD, Skwiat TM, Antonio J (2017) International society of sports nutrition position stand: protein and exercise. *Journal of the International Society of Sports Nutrition* 14: 20. <https://doi.org/10.1186/s12970-017-0177-8>
- Jäger R, Purpura M, Farmer S, Cash HA, Keller D (2018) Probiotic *Bacillus coagulans* GBI-30, 6086 improves protein absorption and utilization. *Probiotics and Antimicrobial Proteins* 10(4): 611–615. <https://doi.org/10.1007/s12602-017-9354-y>
- Johnston AP, Burke DG, MacNeil LG, Candow DG (2009) Effect of creatine supplementation during cast-induced immobilization on the preservation of muscle mass, strength, and endurance. *Journal of Strength and Conditioning Research* 23(1): 116–120. <https://doi.org/10.1519/JSC.0b013e31818efbcc>
- Kantchev D, Nenova G, Naydenov A, Zheleva E, Mihailova-Alakidi V (2024) Application of analytical exercises after anterior cruciate ligament reconstruction for static, dynamic balance and postural control improvement. *Journal of IMAB* 30(3): 5655–5657. <https://doi.org/10.5272/jimab.2024303.5655>
- Kerksick CM, Arent S, Schoenfeld BJ, Stout JR, Campbell B, Wilborn CD, Antonio J (2017) International society of sports nutrition position stand: nutrient timing. *Journal of the International Society of Sports Nutrition* 14: 33. <https://doi.org/10.1186/s12970-017-0189-4>
- Kreider RB, Kalman DS, Antonio J, Ziegenfuss TN, Wildman R, Collins R, Lopez HL (2017) International Society of Sports Nutrition position stand: safety and efficacy of creatine supplementation in exercise, sport, and medicine. *Journal of the International Society of Sports Nutrition* 14: 18. <https://doi.org/10.1186/s12970-017-0173-z>
- Ljungqvist O, Søreide E (2003) Preoperative fasting. *British Journal of Surgery* 90(4): 400–406. <https://doi.org/10.1002/bjs.4066>
- Mihaylova V, Lyochkova M (2012) Salutogenesis: the end of the road or new perspectives? *Social Medicine* 2–3: 18–22.
- Mihaylova-Alakidi V (2019) Current concepts of quality of life; role of physical activity in the aging process. In: *Contemporary dimensions of the problems of successful aging*. Rakursi, Sofia, 69–77, 117–154.
- Molfino A, Gioia G, Rossi Fanelli F, Muscaritoli M (2013) Beta-hydroxy-beta-methylbutyrate supplementation in health and disease: a systematic review of randomized trials. *Amino Acids* 45(6): 1273–1292. <https://doi.org/10.1007/s00726-013-1592-z>
- Nicastro H, Artioli GG, dos Santos Costa A, Solis MY, Da Luz CR, Blachier F, Lancha AH (2011) An overview of the therapeutic effects of leucine supplementation on skeletal muscle under atrophic conditions. *Amino Acids* 40(2): 287–300. <https://doi.org/10.1007/s00726-010-0636-x>
- Nissen SL, Sharp RL (2003) Effect of dietary supplements on lean mass and strength gains with resistance exercise: a meta-analysis. *Journal of Applied Physiology*. <https://doi.org/10.1152/jappphysiol.00755.2002>
- Paddon-Jones D, Sheffield-Moore M, Aarsland A, Wolfe RR, Ferrando AA (2005) Exogenous amino acids stimulate human muscle anabolism without interfering with the response to mixed meal ingestion. *American Journal of Physiology – Endocrinology and Metabolism* 288(4): E761–E767. <https://doi.org/10.1152/ajpendo.00291.2004>
- Paddon-Jones D, Sheffield-Moore M, Urban RJ, Sanford AP, Aarsland A, Wolfe RR, Ferrando AA (2004) Essential amino acid and carbohydrate supplementation ameliorates muscle protein loss in humans during 28 days bedrest. *Journal of Clinical Endocrinology and Metabolism* 89(9): 4351–4358. <https://doi.org/10.1210/jc.2003-032159>
- Parr EB, Camera DM, Areta JL, Burke LM, Phillips SM, Hawley JA, Coffey VG (2014) Alcohol ingestion impairs maximal post-exercise rates of myofibrillar protein synthesis following a single bout of concurrent training. *PLoS ONE* 9(2): e88384. <https://doi.org/10.1371/journal.pone.0088384>
- Phillips SM (2012) Nutrient-rich meat proteins in offsetting age-related muscle loss. *Meat Science* 92(3): 174–178. <https://doi.org/10.1016/j.meatsci.2012.04.027>
- Raverdeau M, Mills KH (2014) Modulation of T cell and innate immune responses by retinoic acid. *Journal of Immunology* 192(7): 2953–2958. <https://doi.org/10.4049/jimmunol.1303245>
- Rayes N, Seehofer D, Theruvath T, Schiller RA, Langrehr JM, Jonas S, Neuhaus P (2005) Supply of pre- and probiotics reduces bacterial infection rates after liver transplantation—a randomized, double-blind trial. *American Journal of Transplantation* 5(1): 125–130. <https://doi.org/10.1111/j.1600-6143.2004.00649.x>
- Reich KA, Chen YW, Thompson PD, Hoffman EP, Clarkson PM (2010) Forty-eight hours of unloading and 24 h of reloading lead to changes in global gene expression patterns related to ubiquitination and oxidative stress in humans. *Journal of Applied Physiology* 109(5): 1404–1415. <https://doi.org/10.1152/jappphysiol.00444.2010>
- Res PT, Groen B, Pennings B, Beelen M, Wallis GA, Gijzen AP, van Loon LJC (2012) Protein ingestion before sleep improves postexercise overnight recovery. *Medicine & Science in Sports & Exercise* 44(8): 1560–1569. <https://doi.org/10.1249/MSS.0b013e31824cc363>
- Schoenfeld BJ, Aragon AA, Krieger JW (2013) The effect of protein timing on muscle strength and hypertrophy: a meta-analysis. *Journal of the International Society of Sports Nutrition* 10: 53. <https://doi.org/10.1186/1550-2783-10-53>
- Smedley F, Bowling T, James M, Stokes E, Goodger C, O'Connor O, Silk D (2004) Randomized clinical trial of the effects of preoperative and postoperative oral nutritional supplements on clinical course and cost of care. *British Journal of Surgery* 91(8): 983–990. <https://doi.org/10.1002/bjs.4578>
- Smith GI, Atherton P, Reeds DN, Mohammed BS, Rankin D, Rennie MJ, Mittendorfer B (2011) Omega-3 polyunsaturated fatty acids augment the muscle protein anabolic response to hyperinsulinaemia-hyperaminoacidaemia in healthy young and middle-aged men and women. *Clinical Science* 121(6): 267–278. <https://doi.org/10.1042/CS20100597>
- Sobolewski EJ, Thompson BJ, Smith AE, Ryan ED (2011) The physiological effects of creatine supplementation on hydration: a review. *American Journal of Lifestyle Medicine* 5(4): 320–327. <https://doi.org/10.1177/1559827611406071>
- Stechmiller JK (2010) Understanding the role of nutrition and wound healing. *Nutrition in Clinical Practice* 25(1): 61–68. <https://doi.org/10.1177/0884533609358997>
- Stout JR, Smith-Ryan AE, Fukuda DH, Kendall KL, Moon JR, Hoffman JR, Mustad VA (2013) Effect of calcium β -hydroxy- β -methylbutyrate (CaHMB) with and without resistance training in men and women 65+ years: a randomized, double-blind pilot trial. *Experimental Gerontology* 48(11): 1303–1310. <https://doi.org/10.1016/j.exger.2013.08.007>
- Tipton KD (2015) Nutritional support for exercise-induced injuries. *Sports Medicine* 45(Suppl 1): 93–104. <https://doi.org/10.1007/s40279-015-0398-4>
- Van Vliet S, Burd NA, Van Loon LJC (2015) The skeletal muscle anabolic response to plant-versus animal-based protein consumption. *Journal of Nutrition* 145(9): 1981–1991. <https://doi.org/10.3945/jn.114.204305>
- Wall BT, Snijders T, Senden JM, Ottenbros CL, Gijzen AP, Verdijk LB, van Loon LJC (2013) Disuse impairs the muscle protein synthetic response to protein ingestion in healthy men. *Journal of Clinical Endocrinology and Metabolism* 98(12): 4872–4881. <https://doi.org/10.1210/jc.2013-2098>